

(Protocol)Aggregating expert-elicited data for prediction: A scoping review of statistical methods, experiments, and applications

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Combining forecasts from experts, compared to a statistical model, is often more accurate when data is sparse or prone to revision. Mathematical approaches for combining forecasts span several disciplines. Economics, marketing, population dynamics, and politics all rely on expert intuition to make predictions about the future. Because expert-elicited combination forecasting is spread across different domains, researchers use a diverse set of definitions to describe the same concepts. The diversity of applications also means methodological approaches are developed in parallel rather than building on one another. Our scoping review aims to unify these varied definitions and methodological approaches, map key terminology, and summarize applications of combination forecasting, experimental designs, and methods to mathematically aggregate expert-elicited data.

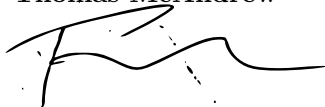
We will collect articles related to methods for aggregating expert-elicited forecasts from the Web of Science database. An article is defined as in-scope if it describes eliciting expert judgments and combining them to make a prediction about a natural phenomena or future event. Articles will be considered in-scope if two committee members read the article and agree it meets the above criteria.

From the final set of in-scope articles, we will extract: the application, experimental design, metrics used to compare forecasts, and the mathematical model used to combine forecasts. Common terminology will be consolidated and discussed. We will also ask of each article a pre-specified set of questions related to combination forecasting.

This protocol prespecifies all criteria and steps for conducting a scoping review on methods for aggregating expert-elicited predictions. All aspects of this protocol were reviewed and agreed on by the coauthors, and any significant deviations from this protocol will be recorded and an amended protocol submitted. This scoping review—selecting articles in a rigorous, prespecified, and unbiased way—aims to map, classify, and summarize all aspects of literature that aggregates expert-elicited predictions.

I. SIGNATURE PAGE

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II. BACKGROUND

Accurate forecasts depend on collecting data related to the target of interest. When data is sparse, or prone to revision, statistical models can struggle to make accurate forecasts of the future. Human judgment is not subject to the same restrictions as statistical models. We can draw on unstructured data, reason about whether past data is predictive of the future, and build complex models of the associations between phenomena. A model combining expert predictions is an attempt to collect this unstructured data and intuition and produce a forecast representing experts' collective knowledge.

This scoping review will map how expert predictions are combined to produce aggregate forecasts. Included articles must have solicited at least two expert human forecasts, and use a mathematical method of aggregating forecasts. Our scoping review will produce simple and concrete definitions, identify approaches to aggregation, experimental designs used to test forecasting model, and expose gaps in the current literature.

In what follows, we describe our scoping review approach. We state our objectives, research question, search strategy, and analysis. This protocol is meant to maintain a rigorous and unbiased strategy for better understanding approaches to combining expert data.

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III. OBJECTIVE

This scoping review’s primary objective is: mapping of key terminology, summarizing applications, experimental designs, and methods to aggregate expert-elicited predictions.

III.1. Primary objectives

1. Map characteristics of aggregating expert data: the timeframe, application, journals where literature is published, methodological approaches, methods of comparison, and experimental designs.
2. Condense a diverse set of terminology into a set of succinct definitions
3. Identify gaps and future directions for this field of research

IV. METHODS

IV.1. Research Question

This scoping review asks “What statistical and quantitative methods are used to aggregate expert-elicited predictions?”. The use of “statistical” and “quantitative” requires scoped articles take a mathematical or statistical approach to learning from expert data. Articles in scope must provide or reference sound statistical techniques.

IV.2. Search strategy

Web of Science will be used to find articles related to combining expert-elicited predictions. We will use the following search term: **(expert* or human* or crowd*) NEAR judgement AND (forecast* or predict*) AND (combin* or assimilat*)** and restrict our search to English-language articles only.

Articles returned from the Web of Science search will be randomly assigned to two reviewers. Each reviewer will read the title and abstract, and determine if the article is in or out of scope. If both reviewers agree the article is considered in scope. If the reviewers disagree, the article will be assigned to a third reviewer to decide

if the article is in or out of scope. We will collect all in scope article’s full text.

V. EXTRACTING INFORMATION FROM FULL TEXT ARTICLES

All in scope full texts will be randomly assigned to a reviewer. The reviewer will read the article in full and answer a set of prespecified questions (Table I). The prespecified questions focus on the experimental design, mathematical/statistical method, sample size, comparison metrics, and application area.

VI. ANALYSIS

A flow chart will be presented that counts the number of articles returned from Web of Science, number of articles after review, and number of final in scope articles. During full-text review, articles may be discovered unrelated to combining expert judgement data and an additional category will be added to the flow-chart “Analysis articles” that counts only those articles used for analysis.

Yes/No prespecified questions will be summarized by frequencies and percentages. The total number of forecasts, and number of expert forecasts will be summarized by mean and standard deviation. If the distribution is skewed or has outliers, the median and IQR will be reported. We will propose a single name that consolidates different ways of referring to the same evaluation metric and model used.

Special attention will be paid to type of model used for aggregating judgements, quality of data judges are thought to use for prediction, and whether or not the model combines point or probability distributions.

VII. PLANS FOR UPDATING THE REVIEW

If the coauthors decide to change the above protocol, an amended protocol will be resubmitted before moving forward and deciding on the final list of included articles for this scoping review. Additional amendments will be labeled $v1.X$, the first amendment labeled $v1.1$, second amendment labeled $v1.2$, and so on.

VIII. STATEMENT OF CONFLICTS OF INTEREST

All authors have no conflicts of interest.

Question	possible answers
Forecasting target	
Identify the primary predictive target?	predictive target
The primary target was categorical	Y/N
The primary target was continuous	Y/N
The primary target was from a time series	Y/N
Experts were given data related to the forecasting target?	Y/N
Terminology	
List terms specific to aggregating crowdsourced data and quoted definition	term,def;term,def
Model	
What models were used in forecasting?	model1,model2, $\dots$, model n
Please list covariates included in any model	cov1,cov2, $\dots$,cov n
A novel model/method was developed	Y/N
Did the authors implement a Bayesian technique?	Y/N
Did the authors use a Frequentist technique?	Y/N
Did the model account for correlation among experts?	Y/N
The model combined point estimates	Y/N
The model combined probabilistic distributions	Y/N
The model was parametric	Y/N
The model was nonparametric	Y/N
Analysis data	
Experts depended on data that could be updated, revised, or rapidly change?	Y/N
Experimental design	
A comparative experiment was conducted	Y/N
How many expert forecasters were included?	integer
How many total forecasts were made?	integer
What evaluation metrics were used?	metric1,metric2, $\dots$,metric n

TABLE I. List of prespecified questions asked of each full-text article. Questions focus on the forecasting target, model, analysis data, experimental design, and future work.